Sequences

A sequence is a list of numbers:

$$\{a_n\}: \quad a_1, a_2, a_3, \ldots$$

Ex. • $\{n\}$ $\quad 1, 2, 3, 4, \ldots \to \infty$ no limit

• $\left\{\frac{n}{n+1}\right\}$ $\quad \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \ldots \to 1$

• $\{(-1)^{n-1}\}$ $\quad 1, -1, 1, -1, 1, -1, \ldots$ no limit

• ? $\quad 3, 1, 4, 1, 5, 9, 2, 6,$ no limit

* These have infinitely many terms!

If $a_n \to L$ as $n \to \infty$, then we say that the sequence $\{a_n\}$ has limit L.

If L exists, we say that $\{a_n\}$ converges.
If not, the sequence diverges.

Ex. Consider the sequence:

$$a_n = \frac{\ln n}{n}.$$

$a_1 = \frac{\ln 1}{1} = 0, \quad a_2 = \frac{\ln 2}{2}, \quad a_3 = \frac{\ln 3}{3}, \quad \ldots$

We check for convergence:

$$\lim_{n\to\infty} \frac{\ln n}{n} \to \text{no l'Hôpital :(}$$

Instead: look at $f(x) = \frac{\ln x}{x}$.
Now we can use l'Hôpital:

$$\lim_{x\to\infty} \frac{\ln x}{x} = \frac{1/x \to 0}{1} = 0.$$

Thus, $\lim a_n = 0$ as well

All the limit rules carry over!
(See the notes for details!)

We define the factorial:

$$n! = n \times (n-1) \times (n-2) \times \cdots \times 3 \times 2 \times 1.$$

e.g. $4! = 4 \times 3 \times 2 \times 1 = 24$

$1! = 1$; $0! = 1$

Ex. Consider $a_n = \dfrac{n!}{n^n}$.

$$a_1 = \frac{1!}{1^1} = 1, \quad a_2 = \frac{2!}{2^2}, \quad a_3 = \frac{3!}{3^3}, \quad \cdots$$

Observation:

$$a_n = \frac{1 \cdot 2 \cdot 3 \cdot 4 \cdots n}{n \cdot n \cdot n \cdot n \cdots n} \begin{array}{l} \leftarrow n \text{ factors} \\ \leftarrow n \text{ times} \end{array}$$

$$= \frac{1}{n} \underbrace{\left(\frac{2 \cdot 3 \cdot 4 \cdots n}{n \cdot n \cdot n \cdots n} \right)}_{\leq 1}$$

$$0 \leq \frac{1}{n} \left(\frac{2 \cdot 3 \cdots n}{n \cdot n \cdots n} \right) \leq \frac{1}{n} \cdot 1$$

as $n \to \infty$ $\downarrow$ 0 $\qquad$ as $n \to \infty$ $\downarrow$ 0

By the Squeeze Theorem: $\lim_{n \to \infty} a_n = 0$.

Series

We construct a series by adding the terms of a sequence:

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots$$

We look at the partial sum:

$$\sum_{i=1}^{n} a_i = a_1 + a_2 + a_3 + \cdots + a_n.$$

To compute the sum of the infinite series, we look at the sequence of partial sums:

$$\lim_{n \to \infty} \sum_{i=1}^{n} a_i$$

If the limit exists, the series converges. Otherwise, the series diverges.

Ex. $a_n = 1$: $1, 1, 1, 1, 1, \ldots$

Does the sequence converge? Yes: 1.

Does the series converge? No.

$$1, \ 1+1, \ 1+1+1, \ 1+1+1+1 \to \infty$$

Ex. The geometric series is given by

$$a + ar + ar^2 + ar^3 + \cdots$$

$$a \xrightarrow{\times r} ar \xrightarrow{\times r} ar^2 \xrightarrow{\times r} ar^3 \cdots$$

Check for convergence: look at the partial sum:

$$S_n = a + ar + ar^2 + \cdots + ar^{n-1}.$$

$$\Rightarrow S_n = \frac{a(1-r^n)}{1-r}.$$

Consider the limit of partial sums:

$$\lim_{n\to\infty} S_n = \lim_{n\to\infty} \frac{a(1-r^n)}{1-r}.$$

This limit exists when $-1<r<1$.

In this case, $r^n \to 0$ so

$$\lim_{n\to\infty} S_n = \frac{a}{1-r}.$$

The geometric series converges if and only if the common ratio r satisfies $-1<r<1$.

Ex. Evaluate $\displaystyle\sum_{n=1}^{\infty} \frac{2^{2n-3}}{6^n}$.

$$\sum_{n=1}^{\infty} \frac{2^{2n-3}}{6^n} = \frac{2^{2-3}}{6} + \frac{2^{4-3}}{6^2} + \frac{2^{6-3}}{6^3} + \cdots$$

Observe that:

$$\frac{2^{2n-3}}{6^n} = \frac{2^{2n}2^{-3}}{6^n} = \frac{1}{8}\cdot\frac{(2^2)^n}{6^n} = \frac{1}{8}\left(\frac{2}{3}\right)^n.$$

Then, $$\sum_{n=1}^{\infty} \frac{1}{8}\left(\frac{2}{3}\right)^n = \frac{1}{8}\cdot\frac{2}{3} + \frac{1}{8}\cdot\left(\frac{2}{3}\right)^2 + \cdots$$

$\times\frac{2}{3}$ $\times\frac{2}{3}$

This is a geometric series:

$$a = \frac{1}{8}\cdot\frac{2}{3} = \frac{1}{12}, \quad r = \frac{2}{3}.$$

Since $-1<\frac{2}{3}<1$, the infinite sum converges:

$$\sum_{n=1}^{\infty} \frac{1}{8}\left(\frac{2}{3}\right)^n = \frac{a}{1-r} = \frac{1/12}{1-2/3} = \frac{1}{4}.$$

Ex. Does the following series converge?

$$\sum_{k=1}^{\infty} \frac{1}{\sqrt{k+3}+\sqrt{k+4}}.$$

Rationalize!

$$\frac{1}{\sqrt{k+3}+\sqrt{k+4}} \times \frac{\sqrt{k+3}-\sqrt{k+4}}{\sqrt{k+3}-\sqrt{k+4}}$$

$$= \frac{\sqrt{k+3}-\sqrt{k+4}}{(k+3)-(k+4)} \leftarrow k-k+3-4$$

$$= \sqrt{k+4}-\sqrt{k+3}.$$

$$\Rightarrow \sum_{k=1}^{\infty} \sqrt{k+4}-\sqrt{k+3}.$$

Look at partial sums:

$$\sum_{k=1}^{n} \sqrt{k+4}-\sqrt{k+3}$$

$$= (\cancel{\sqrt{5}}-\sqrt{4}) + (\cancel{\sqrt{6}}-\cancel{\sqrt{5}}) + (\cancel{\sqrt{7}}-\cancel{\sqrt{6}}) + \cdots + (\sqrt{n+4}-\cancel{\sqrt{n+3}})$$

$$= \sqrt{n+4}-\sqrt{4} = \sqrt{n+4}-2$$

So, $\lim_{n\to\infty} (\sqrt{n+4}-2) \to \infty$.

The series diverges since the partial sums diverge.

Power Series

A power series about $x=a$ has the form

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots.$$

Assume that $a=0$.

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots$$

The convergence of the series depends on x.

Ex. Consider the power series where $c_n = 1$:

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots.$$

This is just a geometric series:

→ converges when $-1 < x < 1$

→ diverges when $x < -1$, $x > 1$.

When $|x| < 1$:

$$1 + x + x^2 + \cdots = \frac{1}{1-x}.$$

Idea: $f(x) = \frac{1}{1-x}$ is just $1+x+x^2+\cdots$ when $-1 < x < 1$.

Q. Which functions can be represented by power series?

$\rightarrow f(x) = \frac{1}{1-x} = 1 + x + x^2 + \cdots \quad [-1 < x < 1]$

Ex. Can we represent $\ln(1+x)$ as a series?

$\star \frac{d}{dx} \ln(1+x) = \frac{1}{1+x} = \frac{1}{1-(-x)}$

$\Rightarrow \frac{d}{dx} \ln(1+x) = 1 + (-x) + (-x)^2 + (-x)^3 + \cdots$

$= 1 - x + x^2 - x^3 + \cdots$

Integrating both sides:

$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + C.$

Find c: plug in $x=0$. Then

$\ln(1) = c \Rightarrow c = 0.$

$\Rightarrow \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$

Let's transform a function $f(x)$ into its power series!

$f(x) = \overset{f(0)}{\cancel{c_0}} + \overset{f'(0)}{\cancel{c_1}} x + c_2 x^2 + c_3 x^3 + \cdots$

Goal: Find $c_0, c_1, c_2, \ldots$

$\rightarrow$ Plug in $x=0$: $f(0) = c_0$.

$\rightarrow$ Differentiate:

$f'(x) = c_1 + 2 \cdot c_2 x + 3 \cdot c_3 x^2 + \cdots$

Plug in $x=0$: $f'(0) = c_1$.

$\rightarrow$ Differentiate again:

$f''(x) = 2 \cdot c_2 + 3 \cdot 2 \cdot c_3 x + \cdots$

Plug in $x=0$: $f''(0) = 2c_2 \Rightarrow c_2 = \frac{f''(0)}{2}$.

$\rightarrow$ Again:

$f'''(x) = 3 \cdot 2 \cdot c_3 + \cdots$

$\Rightarrow f'''(0) = 3 \cdot 2 \cdot c_3 \Rightarrow c_3 = \frac{f'''(0)}{3!}$

$\vdots$

Repeat!

$f^{(n)}(x) = n(n-1)(n-2)\cdots(2)c_n + \cdots$

$\Rightarrow f^{(n)}(0) = n!\, c_n \Rightarrow c_n = \frac{f^{(n)}(0)}{n!}$.

Maclaurin Series

The Maclaurin series of $f(x)$ is given by

$$f(x) = \sum_{n=0}^{\infty} \underbrace{\frac{f^{(n)}(0)}{n!}}_{c_n} x^n$$

$$= f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots$$

<u>Ex.</u> Find the Maclaurin series of

$$f(x) = e^x.$$

For all n, $f^{(n)}(x) = e^x \Rightarrow f^{(n)}(0) = 1$.

Thus,

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots.$$